Identifying Shopping Trends using Data Analysis

A Project Report

submitted in partial fulfillment of the requirements

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AICTE Internship on AI: Transformative Learning
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ABSTRACT

Project Name: Identifying Shopping Trends using Data Analysis

Problem Statement: This project aims to analyze shopping data to identify key trends, customer preferences, and factors influencing purchasing decisions.

Objectives:

- Explore the overall distribution of customer ages.
- Understand how the average purchase amount varies across different product categories.
- Determine which gender has the highest number of purchases.
- Identify the most commonly purchased items in each category.
- Examine how purchase frequency varies across different age groups.

Methodology: The project utilized a dataset containing shopping records, including variables such as age, gender, category, purchase amount, season, review ratings, subscription status, and payment method. Data preprocessing included handling missing values and converting CSV data to Excel. Exploratory data analysis techniques and visualizations, including histograms, bar plots, scatter plots, and pie charts, were employed using pandas, seaborn, and matplotlib.

Key Results & Conclusion: The analysis provided insights into customer demographics, spending patterns, and preferences, helping businesses tailor their marketing strategies, improve customer satisfaction, and optimize inventory management.

TABLE OF CONTENT

LIST OF FIGURES

LIST OF TABLES

CHAPTER 1

Introduction

1.1 Problem Statement:

The problem being addressed is **identifying shopping trends** using data analysis. Gathering data from various sources such as sales records, customer surveys, online shopping behavior, and social media. Cleaning and organizing the data to ensure accuracy and consistency. Using statistical methods and machine learning algorithms to identify patterns, correlations, and trends in the data.

Significance of the Problem

Business Insights: It will help businesses to make informed decisions about inventory management, marketing strategies, and customer engagement.

1.2 Motivation:

This project was chosen because **understanding shopping trends** is crucial for businesses to stay competitive and relevant in today's market. With the rise of e-commerce and the abundance of data available, leveraging data analysis to uncover shopping patterns can provide significant advantages.

Potential Applications

1. **Retail Businesses**: Identify which products are trending and adjust inventory accordingly.

2. **Marketing Teams**: Develop targeted marketing campaigns based on consumer preferences.

Impact

1. **Increased Sales**: By understanding and predicting trends, businesses can better meet customer demands, leading to increased sales.

2. **Customer Loyalty**: Personalized shopping experiences can enhance customer loyalty and retention.

1.3 Objective:

The objective of this project is to delve into the world of consumer shopping behavior by analyzing data to uncover key trends and patterns. By understanding what drives customers' purchasing decisions, we aim to segment them based on their preferences and habits. This insight will not only help businesses predict future trends and demands but also optimize their inventory management and enhance marketing strategies. Ultimately, the goal is to improve customer satisfaction and loyalty, leading to increased revenue and a more personalized shopping experience.

1.4 Scope of the Project:

The scope of this project encompasses the collection, processing, analysis, and visualization of data from various sources such as sales records, customer surveys, online shopping behavior, and social media. By employing statistical methods and machine learning algorithms, the project aims to identify patterns, correlations, and trends in consumer shopping behavior. The findings will be presented through charts, graphs, and dashboards to provide clear and actionable insights. However, the project faces limitations such as the quality and scope of the available data, resource constraints, rapidly changing market conditions

CHAPTER 2

Literature Survey

2.1 Review relevant literature or previous work in this domain.

Researchers have been delving into consumer behavior, purchase preferences, and how discounts or promotions influence sales. Additionally, examining seasonal trends and demographic factors has been key to predicting customer spending patterns.

Advanced tools like machine learning and data visualization are now being used to uncover hidden patterns in transactional data. Clustering techniques have proven effective for segmenting customers, while regression models have been useful in predicting sales trends and understanding the impact of external factors like promotions. These studies underscore the importance of using data to gain insights into consumer behavior and enhance business strategies.

2.2 Mention any existing models, techniques, or methodologies related to the problem.

- **Time Series Analysis**: This method analyzes data points collected or recorded at specific time intervals.
- **Predictive Analytics**: This involves using historical data to make informed predictions about future trends.
- **Data Visualization Tools**: Tools like charts, graphs, and dashboards are used to present data in an easily understandable format.
- **Sentiment Analysis**: This technique analyzes customer reviews, social media posts, and other textual data to gauge customer sentiment.

- **Customer Segmentation Models**: Techniques like K-Means clustering and hierarchical clustering are widely used to group customers based on purchasing behavior and demographics.

2.3 Highlight the gaps or limitations in existing solutions and how your project will address them.

1. **Limited Customization**: Existing models often fail to cater to the specific needs of different industries or businesses, leading to generic insights.

2. **Dynamic Trends**: Many methodologies do not account for rapidly changing consumer preferences and external factors, such as economic conditions or emerging market trends.

3. **Integration of Multidimensional Data**: Most studies focus on isolated factors like demographics or purchase behavior without integrating multiple variables for a holistic analysis.

4. **Seasonal and Temporal Variations**: Existing solutions often overlook the impact of seasons or specific time periods on shopping trends.

Addressing the Gaps

This project aims to bridge these gaps by:

1. Incorporating a multidimensional analysis that considers demographics, seasonal factors, and product attributes simultaneously.

2. Using advanced visualization techniques to present findings in an intuitive and actionable format.

3. Customizing the analysis for specific business questions, ensuring relevance and practical applicability.

CHAPTER 3

Proposed Methodology

3.1 System Design

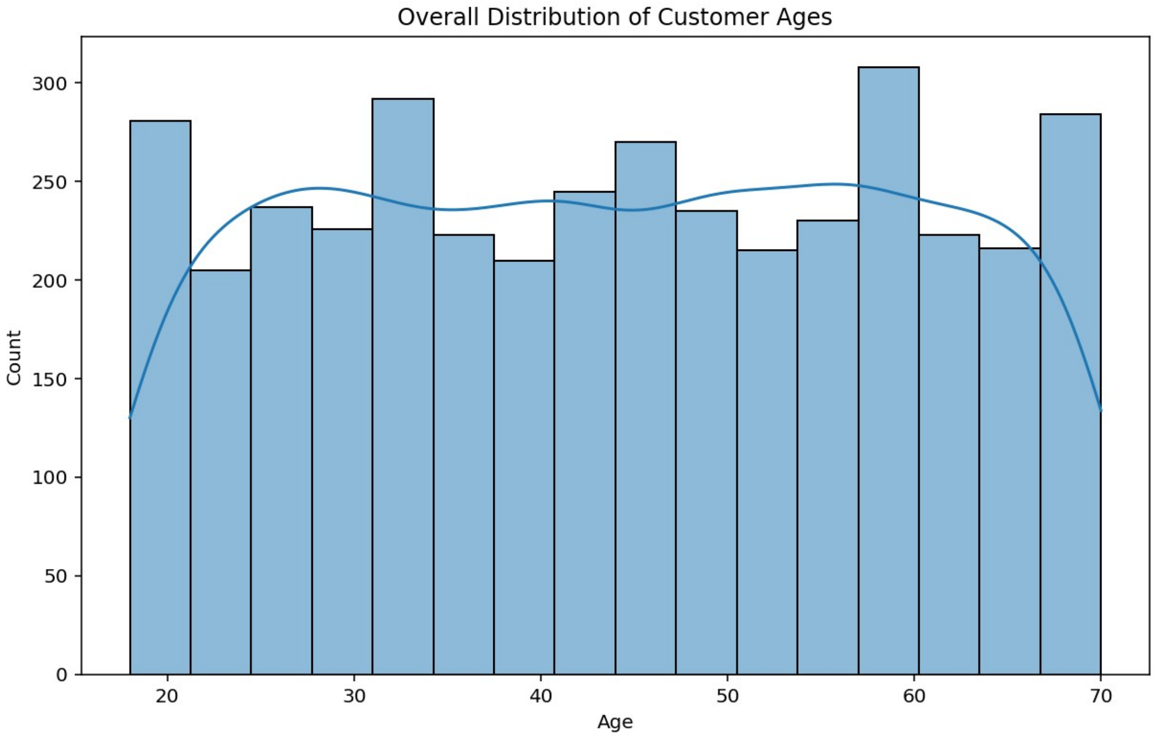

Fig 1.

The histogram **Overall Distribution of Customer Ages** depicts the distribution of customer ages. Firstly, the age range of the customer base spans from approximately 20 to 70 years old. Secondly, the distribution is not uniform, with a noticeable peak around the 30-40 age group, suggesting a concentration of customers in this demographic. The relatively high frequency counts in the 30-40 age bracket imply that this segment constitutes a significant portion of the customer base.

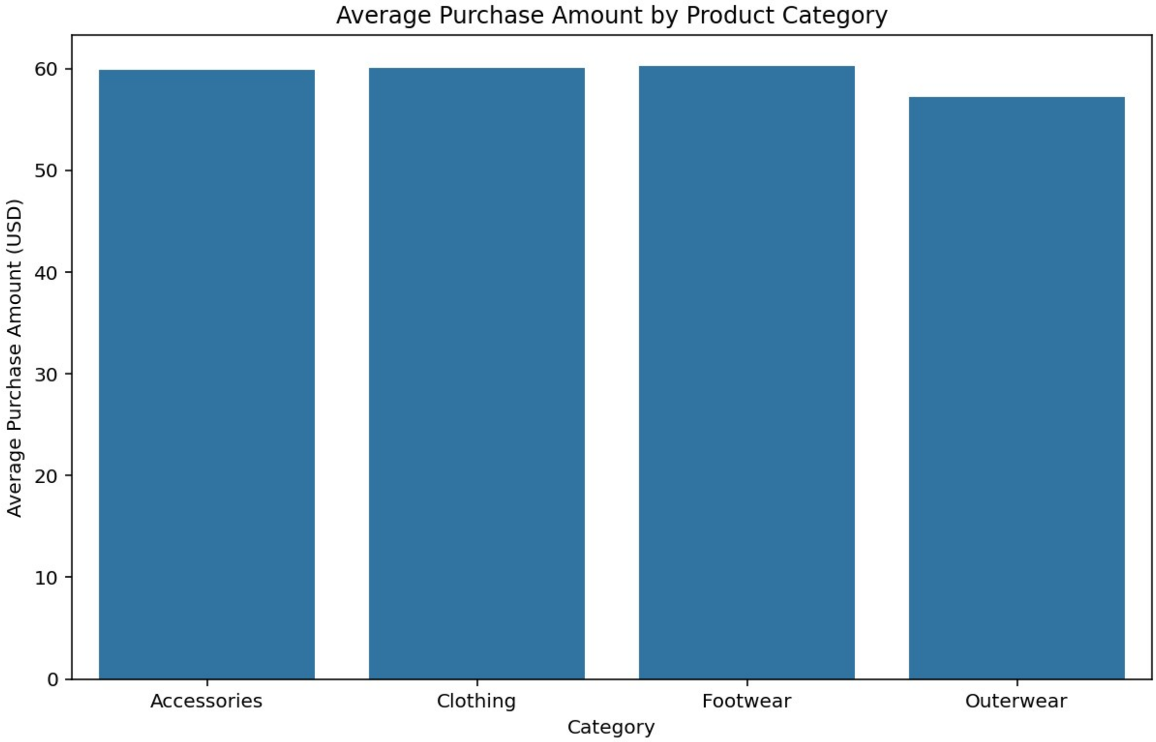

Fig 2.

The bar chart **Average Purchase Amount by Product Category** presents the average purchase amounts for different product categories. It reveals that customers tend to spend the most on Accessories, with the average purchase amount hovering around $60. Clothing and Footwear follow closely behind, both with average purchase amounts slightly lower than Accessories. Outerwear, on the other hand, has the lowest average purchase amount, suggesting that customers might be more price-sensitive or less inclined to spend on this category.

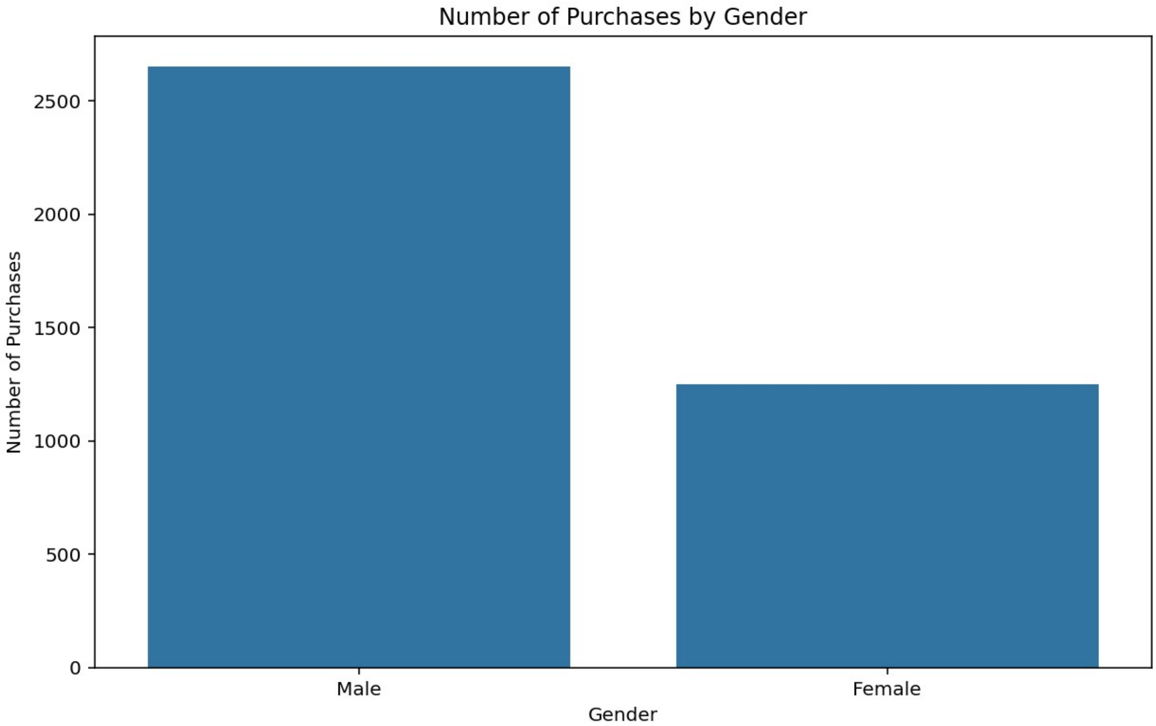

Fig 3.

The bar graph titled "**Number of Purchases by Gender**" presents a stark contrast in purchasing behavior between males and females. The bar representing male purchases is significantly taller than the bar for female purchases, indicating that males made considerably more purchases than females during the period this data covers.

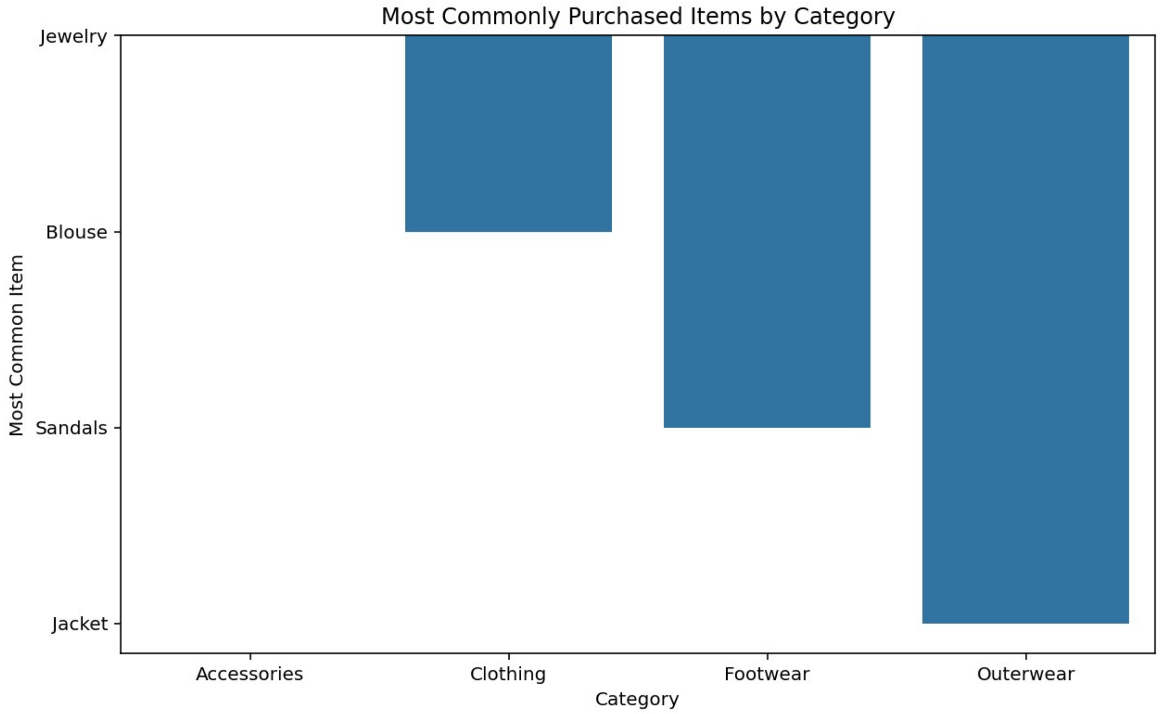

Fig. 4

The Bar Graph titled "**Most commonly purchased item by category**" shows Category on X-axis and Most Common Item on Y-axis. The inference that can be drawn from this graph is that Jacket is the most common item Purchased in Outerwear Category, Sandals is the most common item in Footwear Category whereas Blouse is the most common item Purchased in Clothing Category.

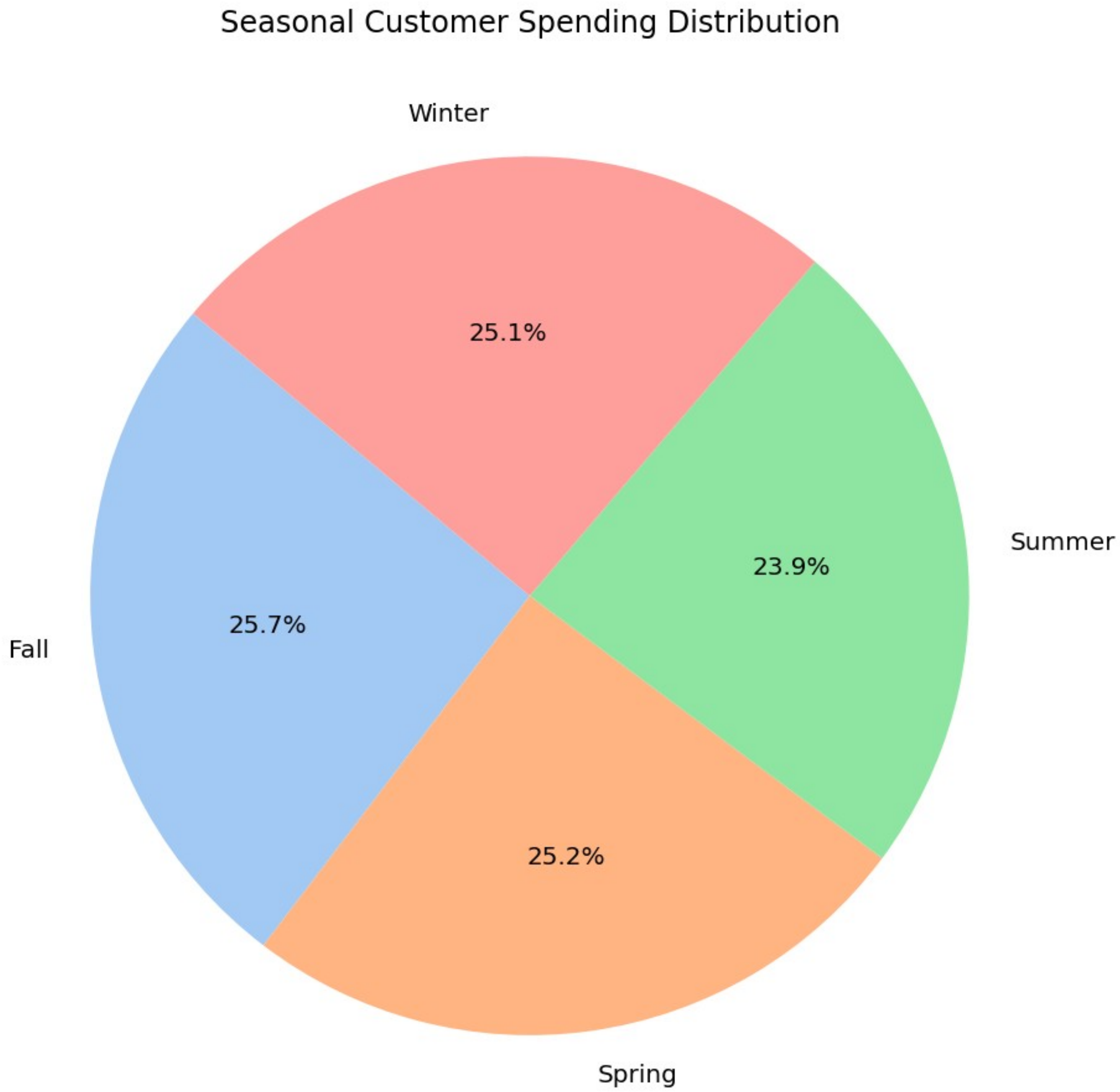

Fig 5.

The pie chart titled **Seasonal Customer Spending Distribution** provides us with the inference that generally customer spending patterns are almost equivalent in all seasons except summer where it is a bit less.

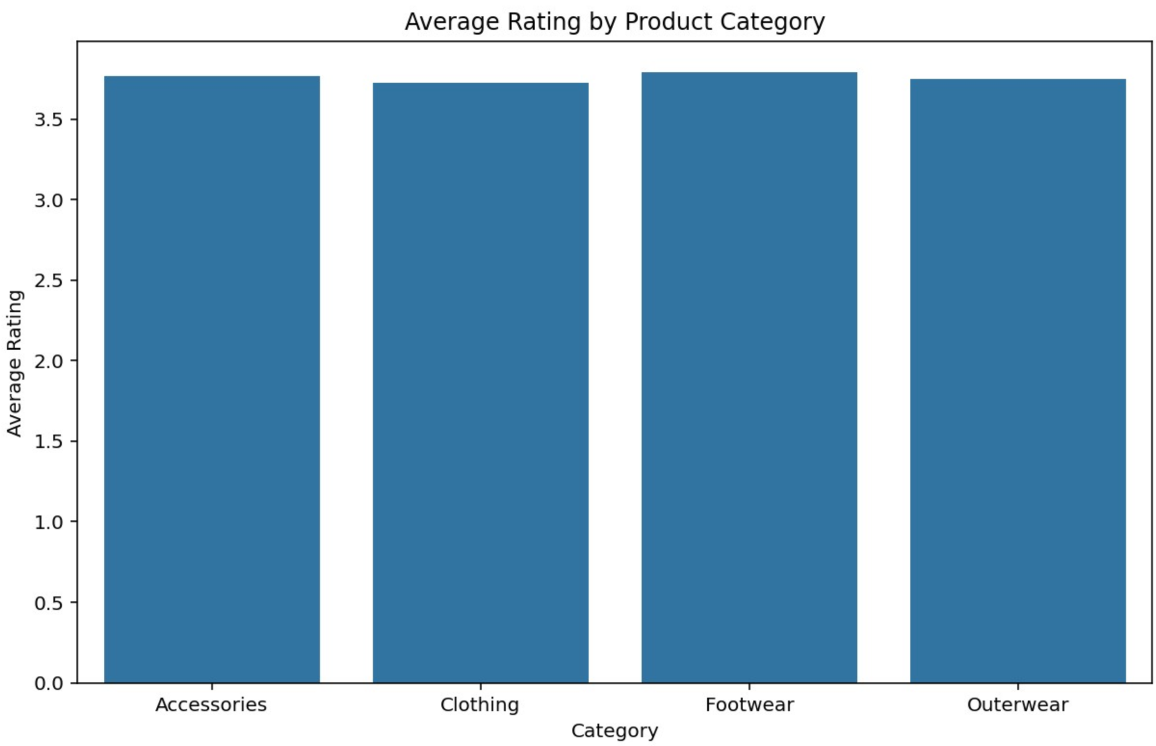

Fig. 6

The bar graph titled "A**verage Rating by Product Category**" presents a picture which shows that the Average Rating of products across different category has been almost identical. The bar representing all the purchases are quite similar and all of them have an average rating of almost 3.5.

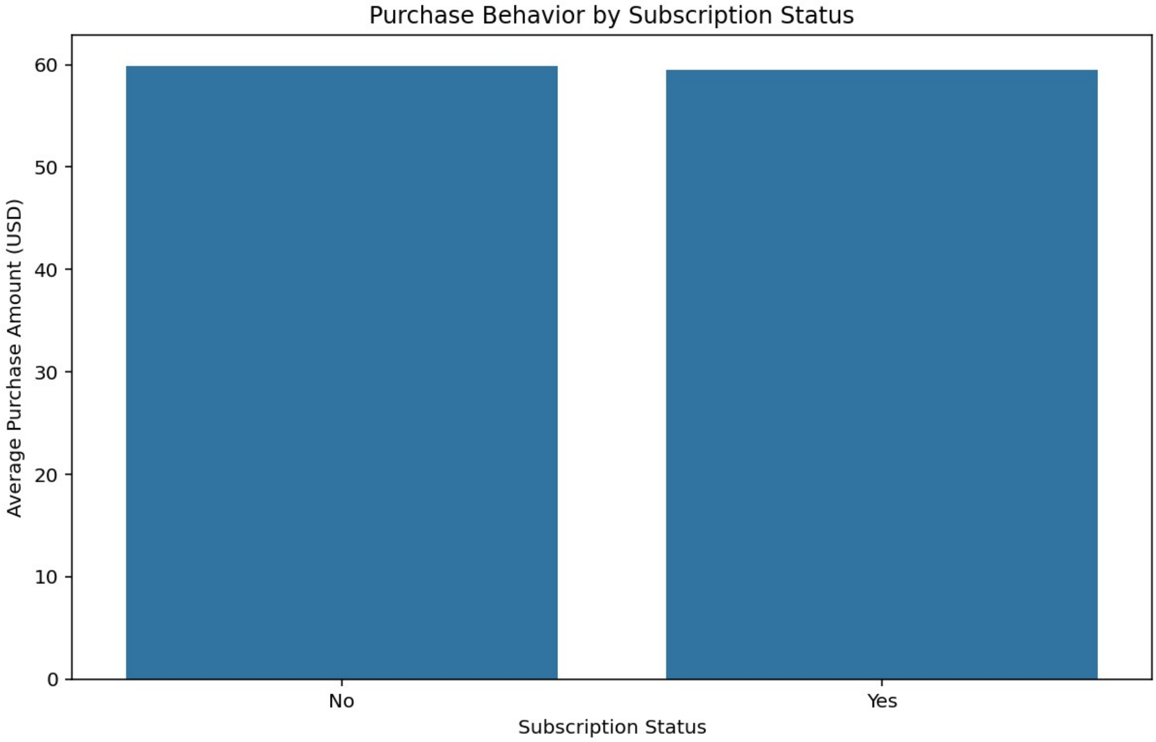

Fig.7

The bar graph titled "**Purchase Behavior by Subscription Status**" shows the average purchase amount for customers with and without subscriptions. Both bars are nearly identical in height, indicating that, on average, customers with and without subscriptions spend roughly the same amount on purchases.

This suggests that the presence or absence of a subscription does not significantly influence a customer's average purchase amount.

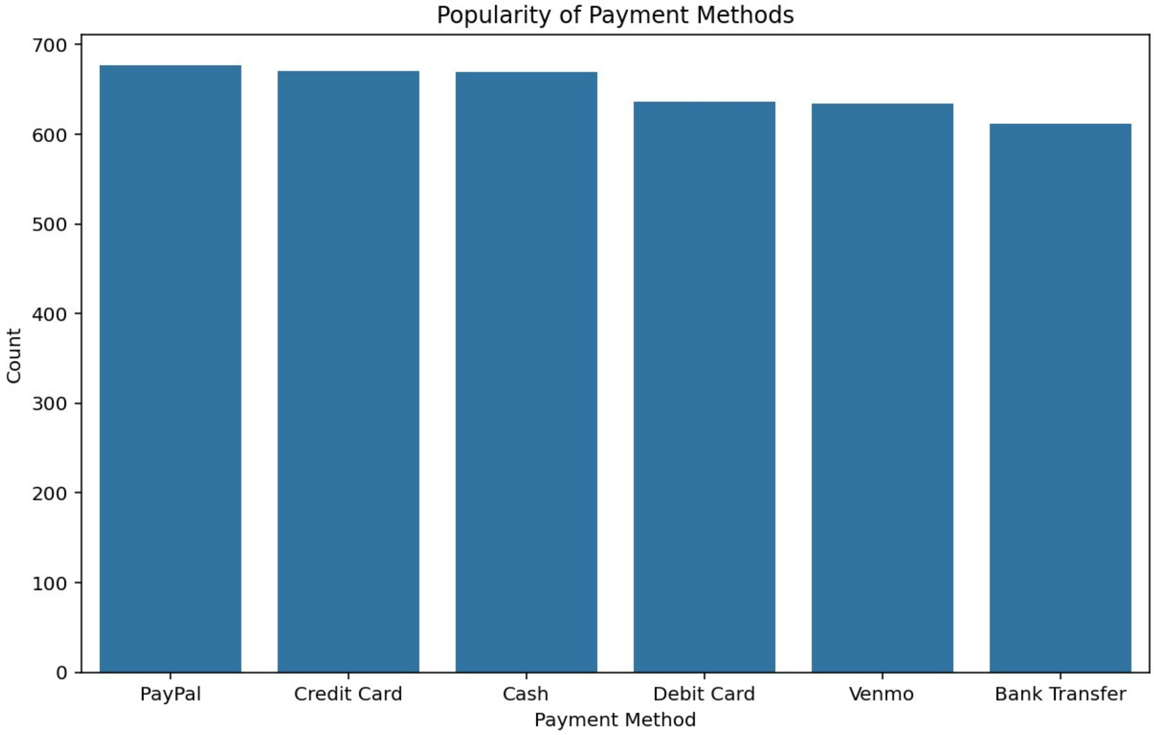

Fig. 8

The bar graph titled "Popularity of Payment Methods" shows the frequency of different payment methods used in a sample of transactions. PayPal is the most popular method, with the highest bar, followed closely by Credit Card and Cash. Debit Card and Venmo have similar usage rates, while Bank Transfer is the least popular method in this dataset.

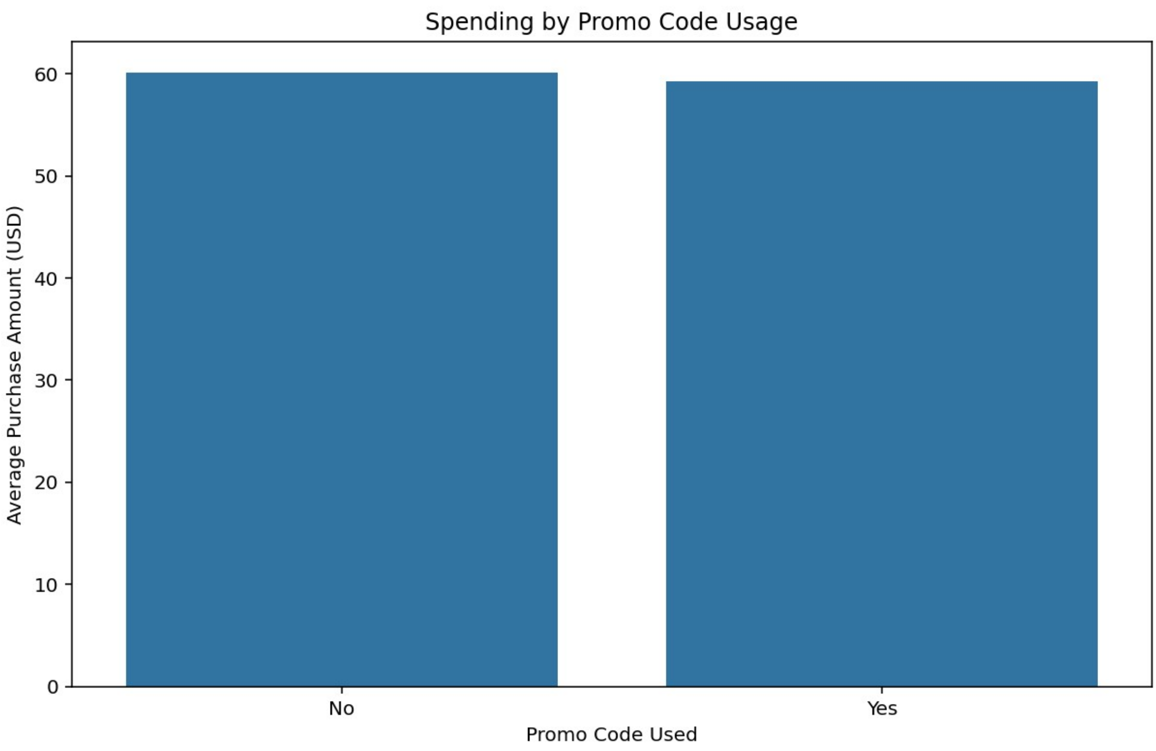

Fig. 9

The bar chart titled “**Spending by Promo Code Usage**” shows the Average Purchase Amount for customers who did and did not use the Promo Code. Interestingly, both groups spent approximately the same amount, around $60, suggesting that the use of promo codes did not significantly impact the average purchase value in this dataset

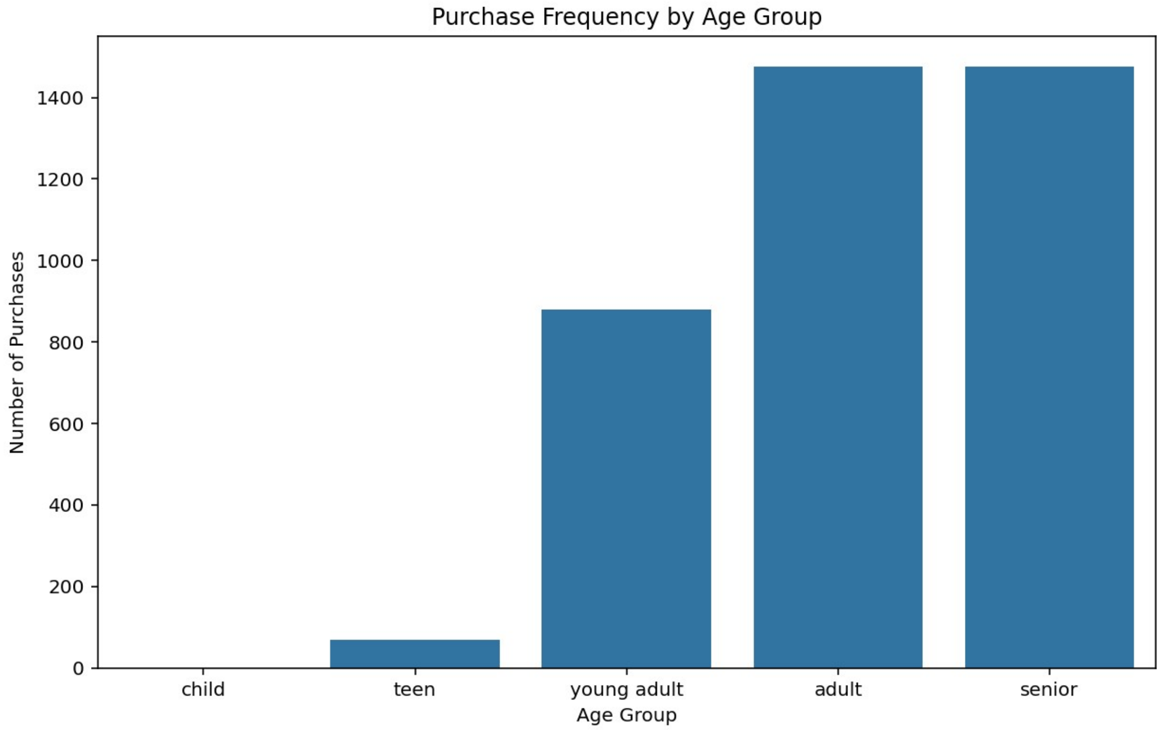

Fig. 10

The bar graph titled "**Purchase Frequency by Age Group**" shows the Number of Purchase Frequency by several different Age group. The basic inference that we can draw from this is that there is no purchase made in the child group, then Teens make purchase between 0-200 , for Young Adult it is 0-800 and among the Adult and Senior it is highest between 0-1400.

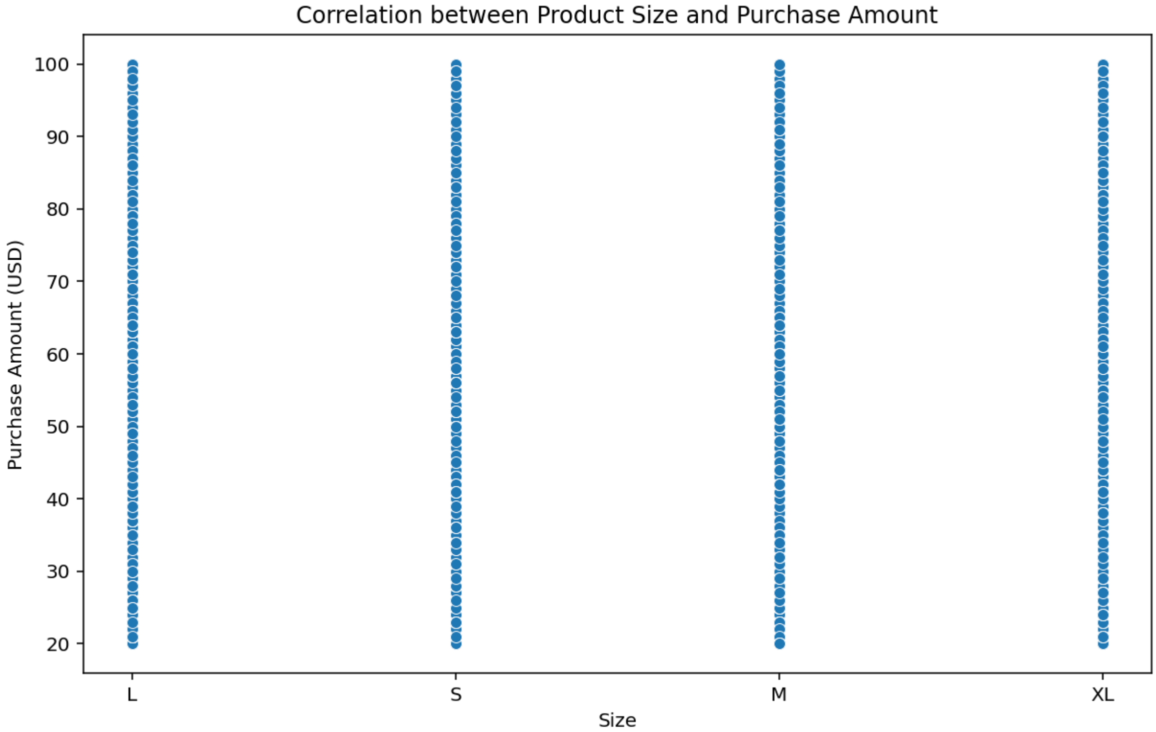

Fig. 11

The Scatter Plot titled "Correlation between Product Size and Purchase Amount" visualizes the relationship between Purchase Amount and Size of a Product(L, S, M, XL). From this plot this inference can be drawn that the purchase amounts within each size category seems to be relatively consistent, with little variations across different product sizes. So it can be said that the product size is not a strong predictor of purchase amount in this dataset.

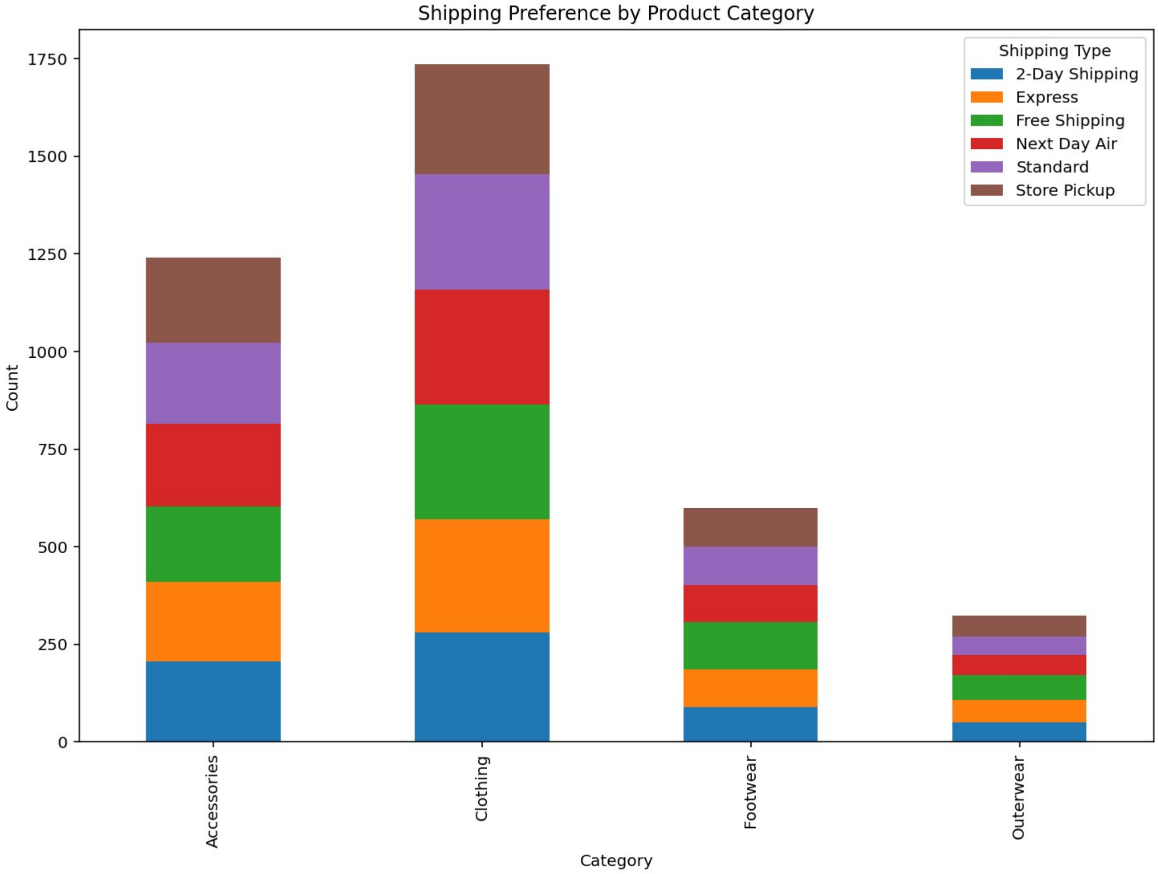

Fig. 12

The stacked bar chart titled "**Shipping Preference by Product Category**" illustrates the distribution of shipping preferences for different product categories: Accessories, Clothing, Footwear and Outerwear. The inference that can be drawn is that **Clothing** has the highest of all orders while **Outerwear** has the lowest and **Standard** shipping is the most popular form of delivery.

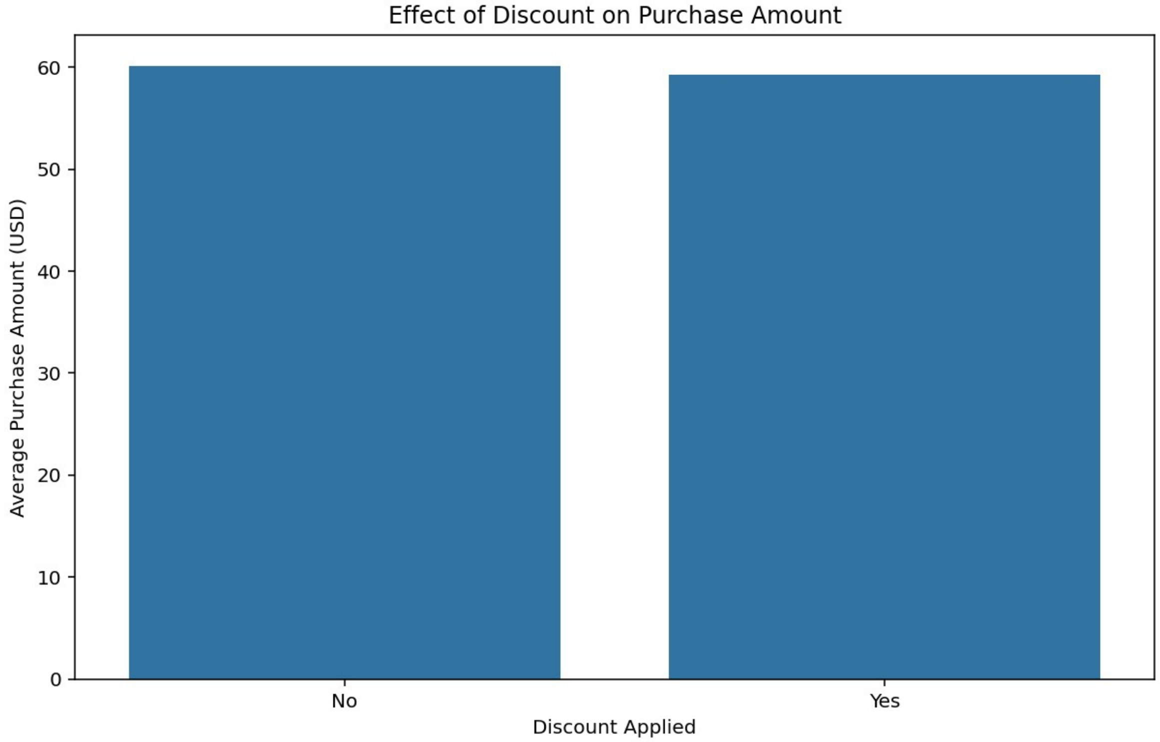

Fig. 13

The bar chart titled "**Effect of Discount on Purchase Amount**" compares the average purchase amount (in USD) for customers who received a discount versus those who did not. Interestingly, both groups spent approximately the same amount, around $60. This suggests that the application of discounts did not significantly impact the average purchase value in this dataset. This finding could be due to various factors, such as the types of discounts offered, the target audience of the discounts, or the overall pricing strategy of the business.

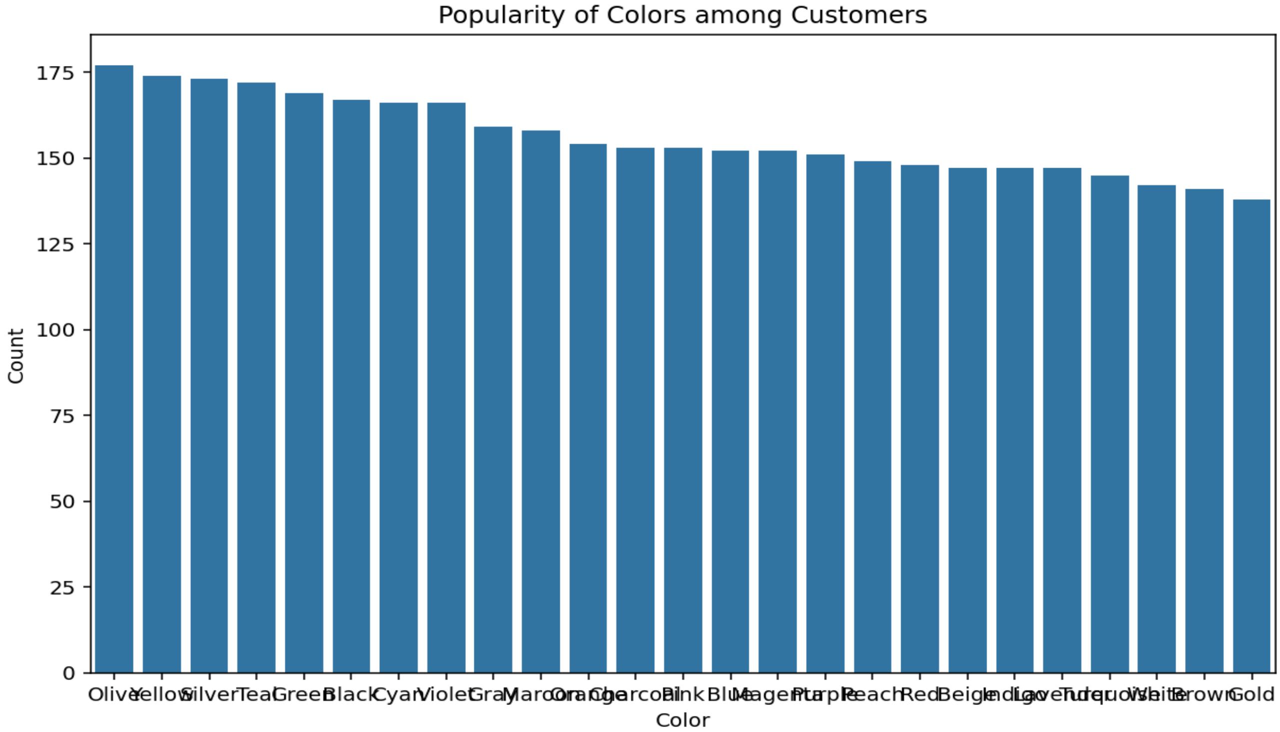

Fig.14

The bar chart titled "**Popularity of Colors among Customers**" shows the number of times each color is selected by customers. Olive is the most popular color with around 175 selections, followed closely by Silver and Teal. Black, Cyan, and Violet are also popular choices, while colors like White, Gold, and Brown have significantly lower selection counts. This suggests that customers have a preference for certain colors over others, with Olive, Silver, and Teal being particularly favored.

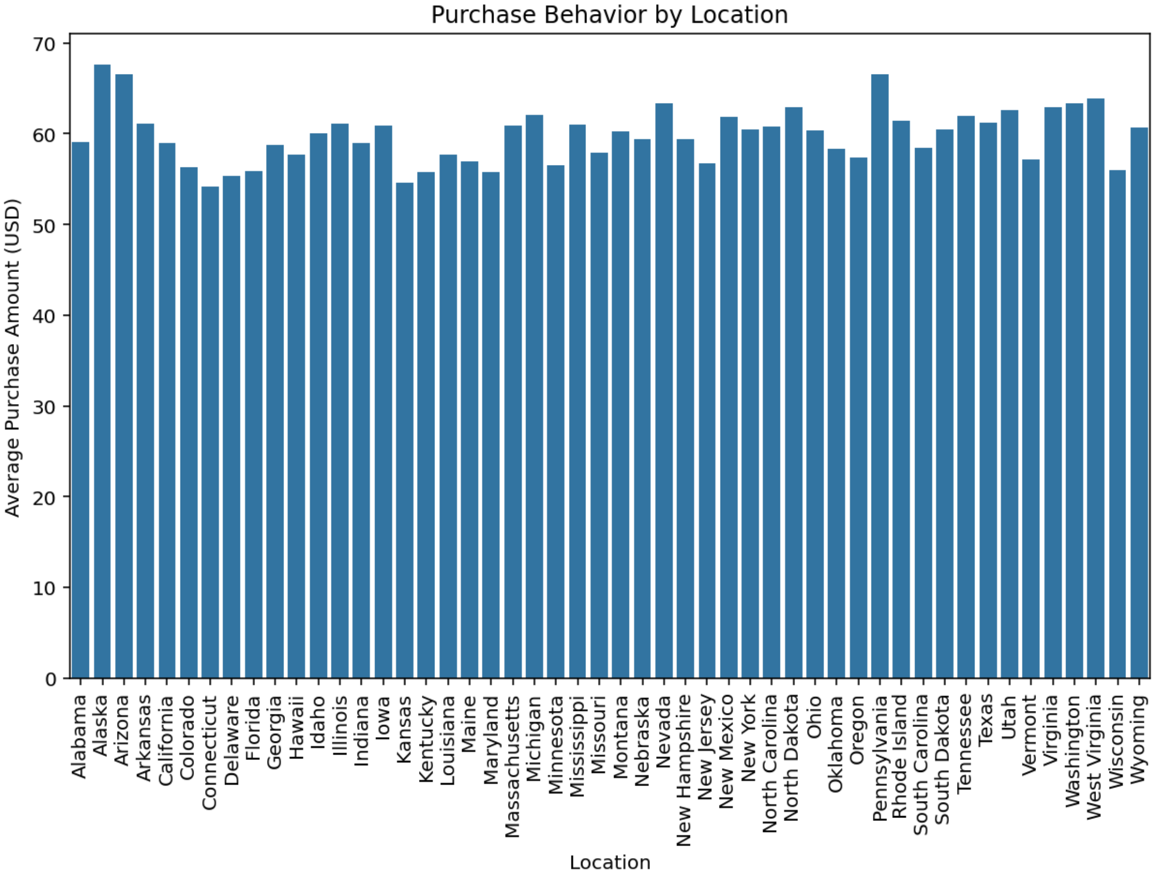

Fig. 15

The bar chart titled "Purchase Behavior by Location" shows the average purchase amount (in USD) for different states in the US. The y-axis represents the average purchase amount, while the x-axis lists all the states. The height of each bar corresponds to the average purchase amount for that particular state. The chart reveals variations in spending habits across different regions of the country. It reveals that Alaska and Pennsylvania are the states with most spendings.

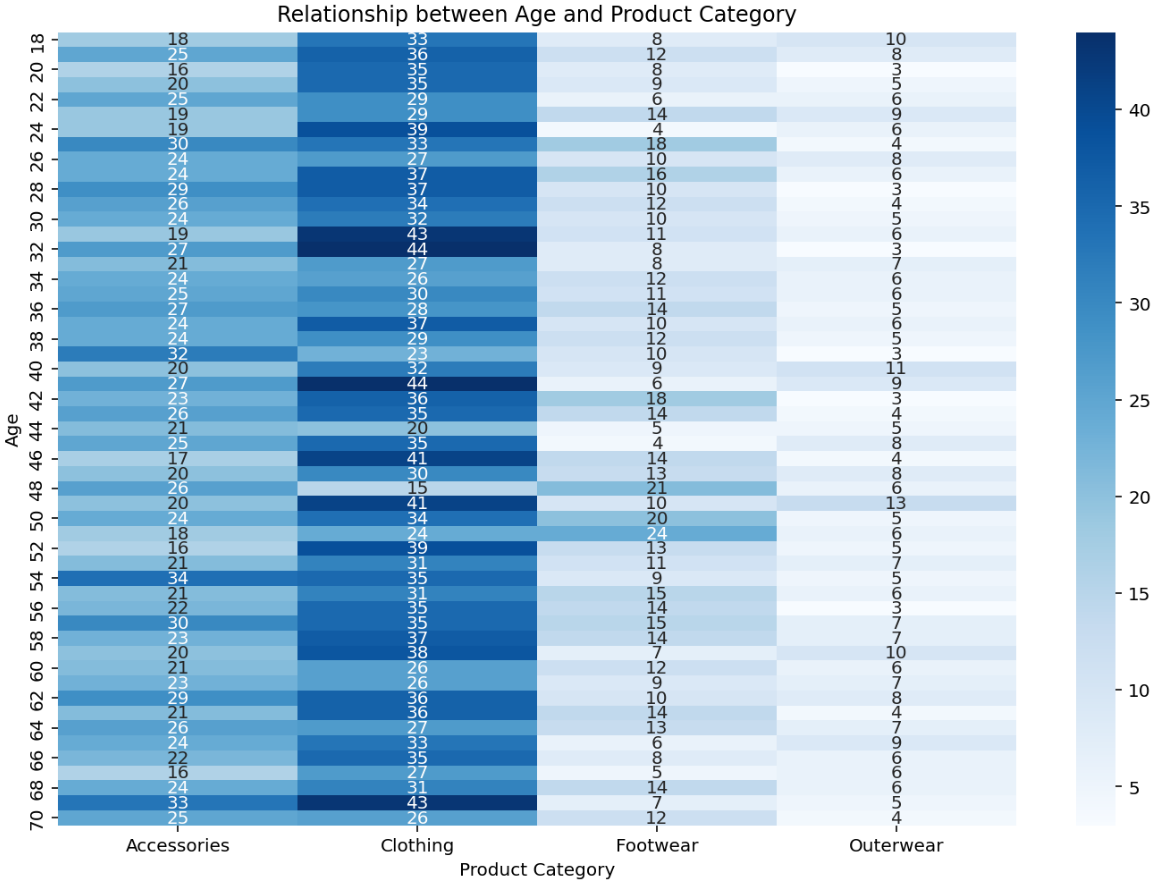

Fig. 16

The heatmap titled "Relationship between Age and Product Category" visualizes the frequency of purchases across different product categories (Accessories, Clothing, Footwear, and Outerwear) for various age groups. The color intensity represents the number of purchases, with darker shades indicating higher frequency. The chart suggests that c younger age groups might show a preference for clothing and accessories, while older age groups might lean towards footwear and outerwear.

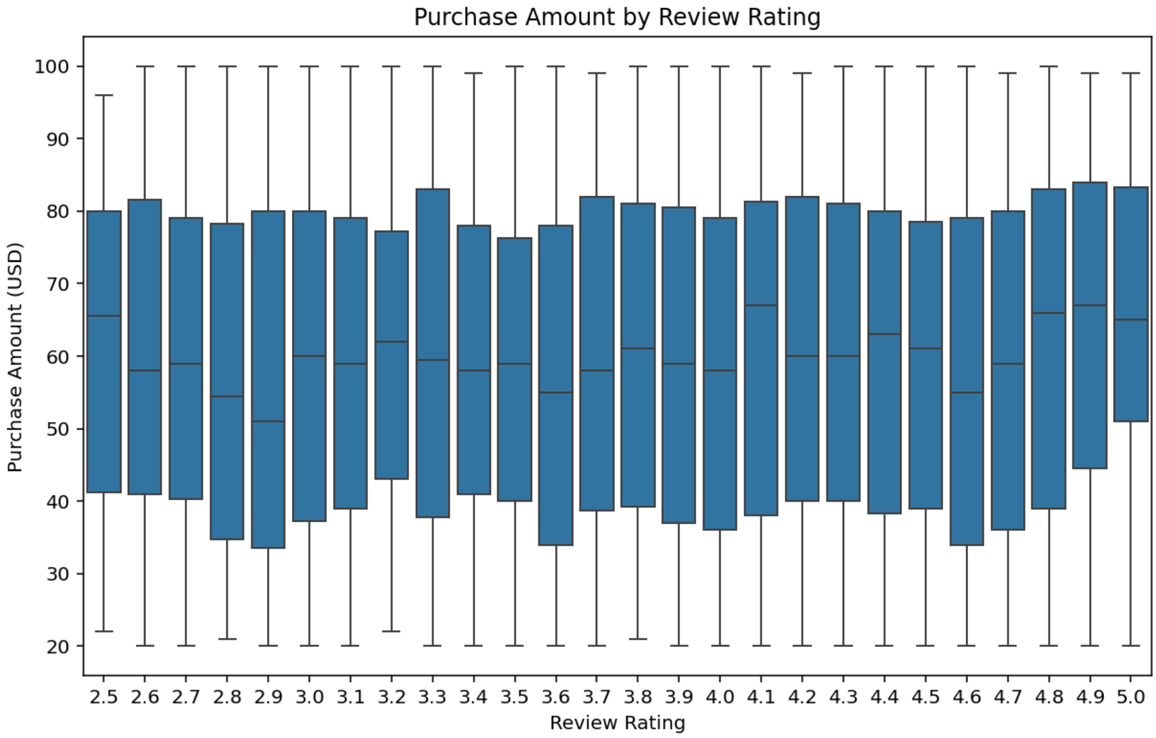

Fig. 17

The Boxplot titled "**Purchase Amount by Review Rating**" shows purchase amounts (in USD) distributed by review ratings. Each box represents the interquartile range (IQR) of purchase amounts for a given review rating, while the horizontal line within each box indicates the median purchase amount. The graph suggests that there is no strong correlation between review ratings and the purchase amount, as the distributions of purchase amounts remain similar regardless of the rating.

3.2Requirement Specification

3.2.1 Hardware Requirements:

These are the hardware resources required to execute the solution:

1. Processor: Minimum Intel Core i5 (or equivalent)

Reason: To handle computations for large datasets efficiently.

2. RAM: 8 GB or more

Reason: Ensures smooth execution of Python scripts and data visualization tools without lag.

3. Storage: At least 500 MB of free disk space

Reason: Required to store the dataset (shopping_trends_updated.csv), converted Excel files, and generated visualizations.

4. Monitor: 1920x1080 resolution (Full HD)

Reason: To properly visualize plots and charts created using seaborn and matplotlib.

3.2.2 Software Requirements:

These are the software tools, libraries, and dependencies required to implement the solution:

1. Operating System:

- **Linux (e.g., Ubuntu 20.04 or later) or Windows 10/11.**

- **Reason: Cross-platform compatibility for running Python scripts and tools.**

2. Python Interpreter:

- **Version 3.8 or later.**

- **Reason: Required for executing the script and utilizing up-to-date libraries.**

3. Libraries and Dependencies:

- **pandas: For data manipulation and analysis.**

- **numpy: For numerical operations and calculations.**

- **seaborn: For creating advanced visualizations like histograms, bar plots, and heatmaps.**

- **matplotlib: For plotting basic graphs and charts.**

4. IDE/Editor:

- **Visual Studio Code or PyCharm**
- **Reason: To write, debug, and execute Python scripts efficiently.**

5. File Formats and Tools:

- **CSV: To load the dataset (shopping_trends_updated.csv).**
- **Excel: For converting data into Excel format using shop.to_excel().**

6. Visualization Tools:

- **Built-in visualization with seaborn and matplotlib.**

7. Dataset Requirements:

- **shopping_trends_updated.csv: A properly formatted dataset containing columns.**
- **Gender, Category, Age, Season, Size, Color, Purchase Amount (USD), Review Rating, etc.**

CHAPTER 4

Implementation and Result

4.1 Snap Shots of Result:

```
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   object
 3   Item Purchased          3900 non-null   object
 4   Category                3900 non-null   object
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                    3900 non-null   object
 8   Color                   3900 non-null   object
 9   Season                  3900 non-null   object
 10  Review Rating           3900 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Shipping Type           3900 non-null   object
 13  Discount Applied        3900 non-null   object
 14  Promo Code Used         3900 non-null   object
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   object
 17  Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

Snap 1

The dataset comprises 3,900 rows and 18 columns, with no missing values across any of the columns. It occupies approximately 548.6 KB of memory.Key columns include `Customer ID`, which uniquely identifies each customer, `Age` for demographic analysis, and `Purchase Amount (USD)` for monetary insights. Other columns, such as `Item Purchased`, `Category`, `Size`, and `Color`, provide details on purchasing trends and product preferences.

```
The unique values of the 'Gender' column are: ['Male' 'Female']

The unique values of the 'Category' column are: ['Clothing' 'Footwear' 'Outerwear' 'Accessories']

The unique values of the 'Size' column are: ['L' 'S' 'M' 'XL']

The unique values of the 'Color' column are: ['Gray' 'Maroon' 'Turquoise' 'White' 'Charcoal' 'Silver' 'Pink' 'Purple'
 'Olive' 'Gold' 'Violet' 'Teal' 'Lavender' 'Black' 'Green' 'Peach' 'Red'
 'Cyan' 'Brown' 'Beige' 'Orange' 'Indigo' 'Yellow' 'Magenta' 'Blue']
```

Snap.2

The image displays the unique values for four different columns in a dataset: "Gender," "Category," "Size," and "Color."

- **Gender: The unique values are "Male" and "Female."**
- **Category: The unique values are "Clothing," "Footwear," "Outerwear," and "Accessories."**
- **Size: The unique values are "L," "S," "M," and "XL."**
- **Color: The unique values include a wide range of colors such as "Gray," "Maroon," "Turquoise," "White," "Charcoal," and many more.**

```
   Customer ID  Age  ... Payment Method Frequency of Purchases
0            1   55  ...          Venmo            Fortnightly
1            2   19  ...           Cash            Fortnightly
2            3   50  ...    Credit Card                 Weekly
3            4   21  ...         PayPal                 Weekly
4            5   45  ...         PayPal               Annually
```

Snap 3

The image contains a table with five rows and four columns. The columns are labeled "Customer ID," "Age," "Payment Method," and "Frequency of Purchases." Each row represents a different customer. The first row shows that customer 1 is 55 years old, pays with Venmo, and makes purchases fortnightly. The second row shows that customer 2 is 19 years old, pays with cash, and also makes purchases fortnightly. The third row shows that customer 3 is 50 years old, pays with a credit card, and makes purchases weekly. The fourth row shows that customer 4 is 21 years old, pays with PayPal, and also makes purchases weekly. Finally, the fifth row shows that customer 5 is 45 years old, pays with PayPal, and makes purchases annually.

4.2 GitHub Link for Code:

https://github.com/Aditya-mj/Data-Analysis-Project

CHAPTER 5

Discussion and Conclusion

5.1 Future Work:

Incorporating advanced feature engineering, such as creating features that capture promotions, holidays, and events. Addressing data sparsity and anomaly detection is crucial. Moving forward, incorporating external data sources, such as weather or social media trends, will provide a more comprehensive view of shopping behaviors. Implementing real-time analysis using tools like Apache Kafka or Spark Streaming will allow me to capture and respond to trends as they happen. If time-related data is available, we could analyze patterns over time to identify trends and make predictions. Additionally, refining the data by handling outliers, studying location-based shopping patterns, and adding more detailed information to the dataset could offer a clearer picture.

5.2 Conclusion:

The project on identifying shopping trends using data analysis has made significant contributions and created a substantial impact in several areas. By leveraging advanced data analysis techniques, the project has provided a deeper understanding of consumer behaviors, uncovering key patterns and trends that drive shopping activities. This insight allows businesses to tailor their strategies more effectively, optimizing inventory management, marketing campaigns, and customer engagement efforts. This not only improves customer satisfaction but also drives increased sales and customer loyalty.

REFERENCES

[1]. https://github.com/phzh1984/Customer-Shopping-Trends-Data-Analysis-and-Visualization

[2]. https://www.youtube.com/playlist?list=PL_iOxYoQ-pUZcqYklbFMR6DgjnzdvTjTK Mentoring Session by Edunet Foundation has been a huge help.

[3]. The code utilizes several libraries and resources for data analysis and visualization. Key references include the Pandas library for data manipulation (Wes McKinney, 2017), NumPy for numerical operations (Travis Oliphant, 2006), Seaborn for statistical visualization (Michael Waskom, 2021), and Matplotlib for plotting (John D. Hunter, 2007).